

The Uniform Walk on a Free Group: Kesten Measure, Exact Excursions, and Radial Escape

HCS-C355 source-local reconstruction

3 September 2026

Abstract

For simple random walk on the rank- d free group, we give a single finite-to-infinite theorem joining operator spectrum, exact returns, and radial escape. With $D = 2d$, the Markov operator has purely absolutely continuous spectrum $[-2\sqrt{D-1}/D, 2\sqrt{D-1}/D]$ and root density $\sqrt{4(D-1)-D^2x^2}/[2\pi(1-x^2)]$. Weighted Dyck decomposition gives closed formulas for every even return and first return, including total return probability $1/(D-1)$. A pathwise boundary-flip coupling then proves speed $(D-2)/D$ and radial central-limit variance $4(D-1)/D^2$. The rank-one boundary becomes the recurrent walk on $\mathbb{Z}$, with arcsine spectrum and half-normal radial scaling. A 1,997-row exact certificate is a regression receipt, not a substitute for the all-time proofs. We claim neither priority nor an arithmetic-target interpretation.

Revision certificate. Round-two escape laws, rank-one boundary, evidence, and Route-A closure.

1 Frozen walk and complete theorem

Let $F_d = \langle a_1, \dots, a_d \rangle$, $d \geq 2$, and put $D = 2d$. Starting from the identity o , multiply on the right at each step by one element of $\{a_i, a_i^{-1} : 1 \leq i \leq d\}$, uniformly. Its Cayley graph is the D -regular tree T_D . On $\ell^2(T_D)$ the transition operator is the bounded self-adjoint operator

$$(Pf)(v) = D^{-1} \sum_{w \sim v} f(w).$$

Write $R_n = |X_n|$ for reduced-word length, and let $u_n = \mathbb{P}_o(X_n = o)$. Set $\rho = 2\sqrt{D-1}/D$ and $C_m = (m+1)^{-1} \binom{2m}{m}$.

Theorem 1 (Spectral, return, and escape atlas). *For the frozen walk:*

(i) P is purely absolutely continuous and $\text{spec}(P) = [-\rho, \rho]$. The spectral measure at o has density

$$k_D(x) = \frac{\sqrt{4(D-1)-D^2x^2}}{2\pi(1-x^2)} \mathbf{1}_{\{|x| < \rho\}}. \quad (1)$$

Equivalently, for $z \notin [-\rho, \rho]$ and the branch $\sqrt{z^2 - \rho^2} \sim z$ at infinity,

$$\langle \delta_o, (z - P)^{-1} \delta_o \rangle = \frac{D\sqrt{z^2 - \rho^2} - (D-2)z}{2(z^2 - 1)}. \quad (2)$$

(ii) $R_0 = 0$, $R_{n+1} = 1$ from $R_n = 0$, while for $r \geq 1$,

$$\mathbb{P}(R_{n+1} = r+1 \mid R_n = r) = p = \frac{D-1}{D}, \quad \mathbb{P}(R_{n+1} = r-1 \mid R_n = r) = q = \frac{1}{D}. \quad (3)$$

All odd returns vanish and, for $n \geq 1$,

$$u_{2n} = D^{-2n} \sum_{k=1}^n \frac{k}{2n-k} \binom{2n-k}{n} D^k (D-1)^{n-k}. \quad (4)$$

(iii) If $\tau_o^+ = \inf\{n \geq 1 : X_n = o\}$, then

$$\mathbb{P}_o(\tau_o^+ = 2k) = \frac{C_{k-1}(D-1)^{k-1}}{D^{2k-1}}, \quad \mathbb{P}_o(\tau_o^+ < \infty) = \frac{1}{D-1}. \quad (5)$$

(iv) With $v = (D-2)/D$ and $\sigma^2 = 4(D-1)/D^2$,

$$\frac{R_n}{n} \longrightarrow v \quad \text{a.s.}, \quad \frac{R_n - vn}{\sqrt{n}} \Longrightarrow N(0, \sigma^2). \quad (6)$$

(v) At the excluded rank-one boundary $d = 1$, $F_1 \cong \mathbb{Z}$: P is purely absolutely continuous on $[-1, 1]$ with arcsine density, $u_{2n} = 4^{-n} \binom{2n}{n}$, return is recurrent, $R_n/n \rightarrow 0$ a.s., and $R_n/\sqrt{n} \Rightarrow |N(0, 1)|$.

2 Cavity resolvent and the root measure

Delete the edge from a vertex to its parent and let $h(z)$ be the diagonal resolvent of the resulting rooted forward tree. Schur complementation at the root of that tree and then at o gives

$$h = \frac{1}{z - (D-1)h/D^2}, \quad G_o(z) = \frac{1}{z - h/D}. \quad (7)$$

The solution satisfying $h(z) \sim z^{-1}$ is

$$h(z) = \frac{D^2}{2(D-1)}(z - \sqrt{z^2 - \rho^2}).$$

Substitution and rationalization prove (2). For $|x| < \rho$, the upper boundary value has

$$-\pi^{-1} \operatorname{Im} G_o(x + i0) = \frac{D\sqrt{\rho^2 - x^2}}{2\pi(1 - x^2)} = k_D(x).$$

The apparent singularities at $z = \pm 1$ in (2) are removable: the numerator vanishes there with the branch fixed above. There are therefore no exterior atoms. Stieltjes inversion gives (1) on the open band. Its integral is one (substitute $x = \rho \cos \theta$ and evaluate the resulting rational sine integral), while the square-root endpoints have no poles. Thus no singular-continuous or endpoint mass remains. The independent asymptotic check $zG_o(z) \rightarrow 1$ confirms the total normalization. Hence (1) is the full probability measure of the cyclic root vector in the radial subspace.

3 Weighted Dyck paths and exact returns

Distance from o changes by one. From a positive radius, precisely one letter cancels the last reduced letter and $D-1$ extend it; at radius zero all D letters lead to radius one. This proves (3). A return of length $2n$ is thus a Dyck path with n rises and n falls. Suppose it has exactly k irreducible excursions from zero. Its k rises from zero have D choices each; the remaining $n-k$ rises have $D-1$ choices, and every fall is forced.

It remains to count the paths. If $C(t) = 1 + tC(t)^2$ is the Catalan series, the required number is $[t^{n-k}]C(t)^k$. Lagrange inversion gives

$$[t^{n-k}]C(t)^k = \frac{k}{2n-k} \binom{2n-k}{n}. \quad (8)$$

Multiplying (8) by $D^k(D-1)^{n-k}$, summing over k , and dividing by D^{2n} proves (4). This also proves the renewal identity $u_{2n} = \sum_{k=1}^n \mathbb{P}(\tau_o^+ = 2k)u_{2n-2k}$ directly at word-count level.

4 Why the whole operator is purely absolutely continuous

The root measure alone controls only radial functions, so we record the complete orthogonal decomposition. Let S_m be the sphere of radius m and let $\phi_m = |S_m|^{-1/2} \mathbf{1}_{S_m}$. The radial subspace generated by ϕ_0 reduces P and has Jacobi matrix

$$J_0 = J(0; a_0, a, a, \dots), \quad a_0 = D^{-1/2}, \quad a = \frac{\sqrt{D-1}}{D}. \quad (9)$$

It is cyclic at its first coordinate, so (1)–(2), including the absence of singular mass, show that J_0 is purely absolutely continuous on $[-2a, 2a]$.

For every vertex v , let $\operatorname{Ch}(v)$ be its children relative to o , and choose an orthonormal basis of the zero-sum space

$$E_v = \{\eta \in \ell^2(\operatorname{Ch}(v)) : \sum_{w \in \operatorname{Ch}(v)} \eta(w) = 0\}.$$

Its dimension is $D-1$ when $v = o$ and $D-2$ otherwise. Given $\eta \in E_v$, let $e_j^{v,\eta}$, $j \geq 0$, assign to every descendant at depth j below $\operatorname{Ch}(v)$ the value of its first child divided by $(D-1)^{j/2}$, and vanish elsewhere. These vectors are orthonormal, the zero-sum condition cancels the edge back to v , and direct substitution gives

$$Pe_0^{v,\eta} = ae_1^{v,\eta}, \quad Pe_j^{v,\eta} = a(e_{j-1}^{v,\eta} + e_{j+1}^{v,\eta}) \quad (j \geq 1). \quad (10)$$

For clarity, $e_0^{v,\eta} = \eta$ and the displayed normalization at depth j uses $(D-1)^{-j/2}$.

These half-line spaces and the radial space are mutually orthogonal. They are complete: on every finite rooted truncation, recursively split the values on each sibling set into its mean and zero-sum parts; induction on the outermost occupied level exhausts every finitely supported vector. Density then exhausts $\ell^2(T_D)$. Thus P is an orthogonal direct sum of J_0 and copies of the free half-line Jacobi matrix $J(0; a, a, \dots)$. The latter is carried by the sine transform to multiplication by $2a \cos \theta$ on $L^2(0, \pi)$, hence is purely absolutely continuous with spectrum $[-2a, 2a]$. This proves the full-space assertion in Theorem 1(i), without extrapolating from one spectral measure.

5 First return and transience

A first return at time $2k$ is one irreducible Dyck excursion. Removing its first rise and last fall leaves one of C_{k-1} Dyck paths. The first rise has D choices, the other $k-1$ rises have $D-1$ choices, and falls are forced. Hence the word count is $DC_{k-1}(D-1)^{k-1}$, proving the first formula in (5). With $p = (D-1)/D$ and $q = 1/D$, the Catalan generating function yields

$$\sum_{k \geq 1} C_{k-1} p^{k-1} q^k = \frac{1 - \sqrt{1 - 4pq}}{2p} = \frac{q}{p} = \frac{1}{D-1},$$

because $p > q$. The missing mass is escape, not an uncounted finite return.

6 Escape strong law and central limit theorem

The boundary at zero requires an explicit coupling. Let $\xi_1, \xi_2, \dots$ be i.i.d. with $\mathbb{P}(\xi_i = 1) = p$ and $\mathbb{P}(\xi_i = -1) = q$, and recursively set

$$\hat{R}_{n+1} = \hat{R}_n + \xi_{n+1} + 2\mathbf{1}_{\{\hat{R}_n=0, \xi_{n+1}=-1\}}, \quad \hat{R}_0 = 0. \quad (11)$$

This chain has exactly (3), so it has the law of R . From radius one the probability of hitting zero is q/p : solve the bounded gambler's-ruin recurrence $h_r = ph_{r+1} + qh_{r-1}$ with $h_0 = 1$ and $h_r \rightarrow 0$. After every departure from zero, the strong Markov property therefore gives fixed escape probability $1 - q/p > 0$. The number of returns, and hence the number

$$B_\infty = \sum_{n \geq 0} \mathbf{1}_{\{\hat{R}_n=0, \xi_{n+1}=-1\}},$$

is finite almost surely. Iterating (11) gives

$$\hat{R}_n = \sum_{i=1}^n \xi_i + 2B_n, \quad B_n \uparrow B_\infty < \infty. \quad (12)$$

The i.i.d. strong law and central limit theorem now apply to the first term; the finite correction divided by n or $\sqrt{n}$ vanishes. Since $\mathbb{E}\xi_i = p - q = (D-2)/D$ and $\text{Var } \xi_i = 4pq = 4(D-1)/D^2$, (6) follows. Equation (12) is also why no conditioning-on-the-last-return gap is hidden in the CLT.

7 Rank-one boundary, exact receipt, and scope

When $d = 1$ and $D = 2$, the Cayley graph is $\mathbb{Z}$. Fourier transform takes P to multiplication by $\cos \theta$, giving the arcsine density $[\pi\sqrt{1-x^2}]^{-1}$ on $[-1, 1]$. Direct word counting gives $u_{2n} = 4^{-n} \binom{2n}{n}$; its divergence criterion or the symmetric gambler's-ruin limit gives recurrence. Here R_n has the law of the absolute value of symmetric walk, so the strong law and invariance principle give $R_n/n \rightarrow 0$ a.s. and $R_n/\sqrt{n} \Rightarrow |N(0, 1)|$. Thus (6) is not silently continued through the critical boundary.

The exact receipt uses $D \in \{4, 6, 8, 10\}$: 1,156 radial-DP rows through time 32, 260 return rows, 256 first-return rows, 256 renewal rows, four parameter rows, and 65 rank-one rows, totaling 1,997 rows and 9,483 scalar cells. An independent implementation checks every row; 237 symbolic identities, two isolated byte replays, and 74 hostile mutations form separate gates. Finite computation proves none of the pure-AC, infinite-sum, strong-law, or CLT statements; the preceding arguments do.

Kesten's original article establishes the symmetric group-walk and free-group spectral-radius lineage [1]; the AMS DOI and JSTOR DOI listed below identify that same article, not two independent sources. Woess provides an authoritative graph-and-group random-walk monograph lineage [2]. We make no priority claim; the contribution here is a source-local, normalization-locked reconstruction joining the results and their boundaries.

The Route-A tuple is

$$(A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_FORMAL_HINT).$$

The last entry records only that P is a natural self-adjoint operator. The rank and clock have no structural arithmetic origin, closed random words are not primitive arithmetic orbits, and (2) is not a target determinant. There is no target Euler product, divisor, functional equation, counting law, zero match, or Hilbert–Pólya claim. Route A is rejected and Route B is not invoked.

References

- [1] H. Kesten, "Symmetric random walks on groups," *Trans. Amer. Math. Soc.* 92 (1959), 336–354. doi:10.1090/S0002-9947-1959-0109367-6; same record at doi:10.2307/1993160.
- [2] W. Woess, *Random Walks on Infinite Graphs and Groups*, Cambridge Tracts in Mathematics 138, Cambridge University Press, 2000. doi:10.1017/CBO9780511470967.